Evolution to 5G-Advanced

Powered by
Nokia Technology Training & Certifications

Evolution to 5G-Advanced

Module Structure

1. 5G-Advanced Key Areas
2. Radio Performance Boosters
3. AI/ML
4. Extended and Augmented Reality
5. Aerial and Space
6. Superaccurate Positioning
7. RedCap for IoT
8. Energy Efficiency

5G-Advanced Overview

1

Evolution

5G-Advanced evolves the 5G system to its fullest capabilities

2

Backwards Compatibility

5G-Advanced maintains backward compatibility to 5G

3

Timing

First version of 5G-Advanced standards in the beginning /mid of 2024.

4

Myths

5G-Advanced denominates the collection of enhancements from Rel 18 onward, it is not defined as one main feature

5

Market longevity

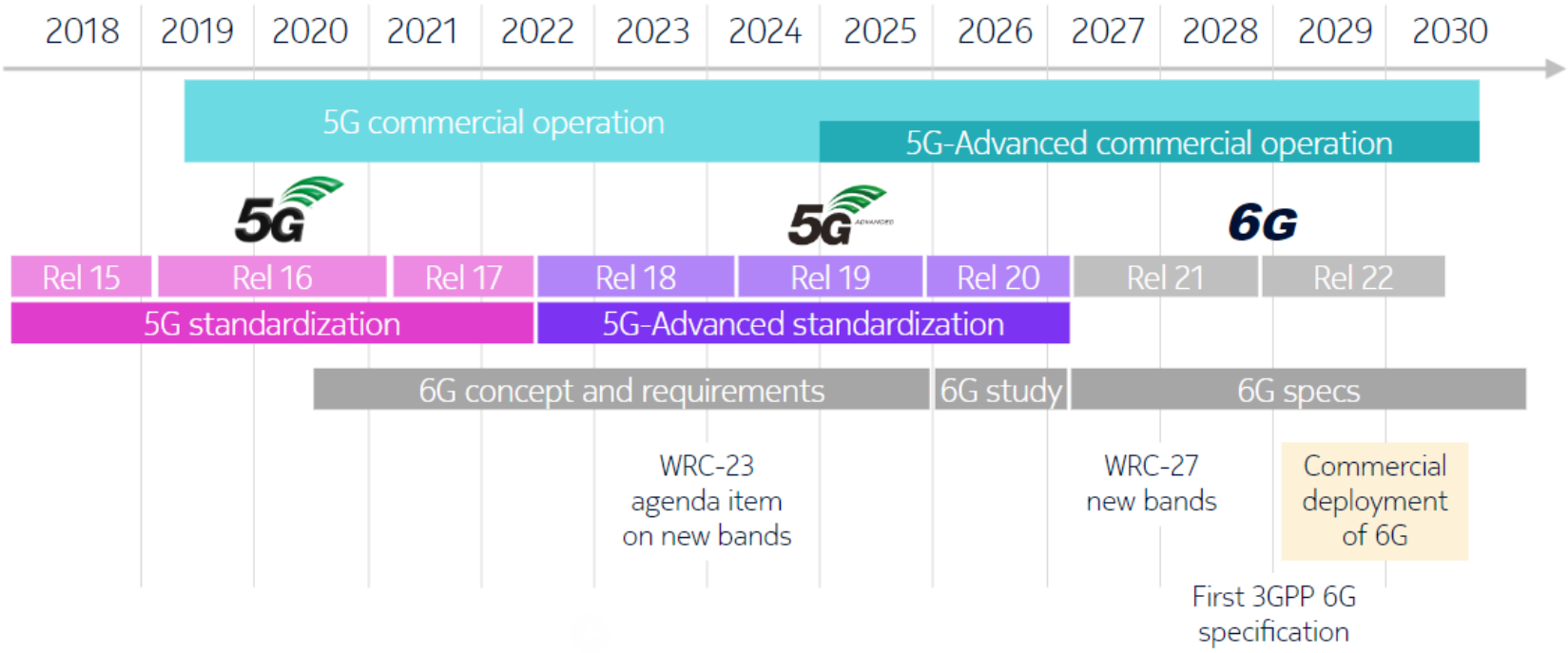
5G-Advanced will have min 5 years of market opportunity.

6

G Revolution

5G-Advanced sets foundation and will be followed by 6G

5G and 5G-Advanced determine network capabilities until 2030



WRC : World Radiocommunication Conferences

5G-Advanced

New usage areas, new services, boosted resiliency and operability

Extreme MBB experience with bounded latency, high date rates enabling IIoT for FR2

High speed mobility and edge roaming

Indoor XR/VR

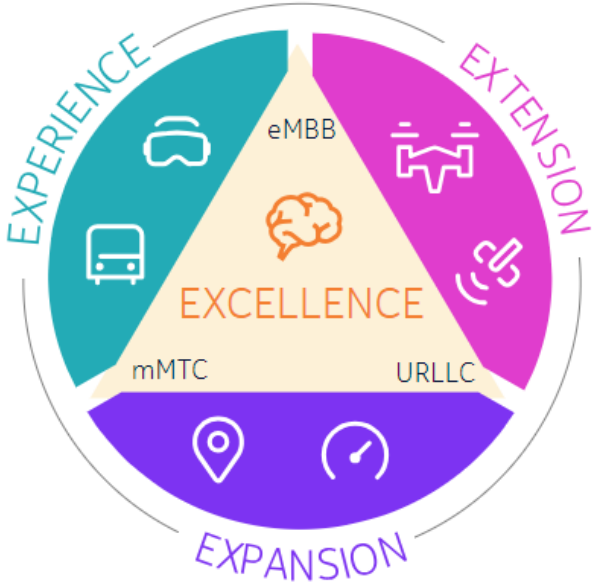
Outdoor XR

Expansion to Support Positioning, Time Sync aaS also without GNSS – Enabling new use cases

GNSS alternative & resiliency solution

PoS, banking

Outdoor XR



Extending global 5G-Advanced reach for IoT and basic MBB

UAV traffic, cameras

NTN, HAPS

Railway, power, wearables, sensors, ...

Network-wide AI/ML enhancements, further network energy saving, network slicing

Dynamic & optimized traffic allocation

Zero-touch orches-trator

Energy saving

GNSS Global Navigation Satellite System
HAPS High Altitude Platform Systems
IIoT Industrial IoT
NTN Non-Terrestrial Network
PoS Point of Sale
UAV Uncrewed Aerial Vehicle
VR Virtual reality
XR Extended reality

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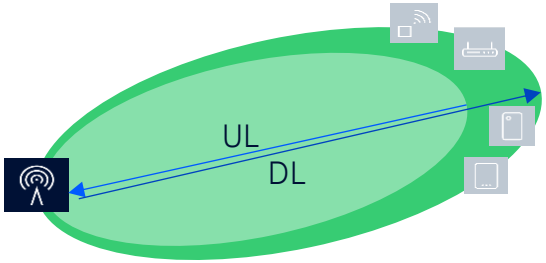
NR Coverage Enhancements



The uplink coverage is more challenging at cell edge than downlink.

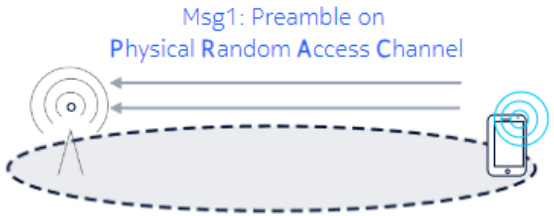
Goal: extend the reach of the network for better uplink coverage.

PRACH Repetitions

Dynamic waveform switching

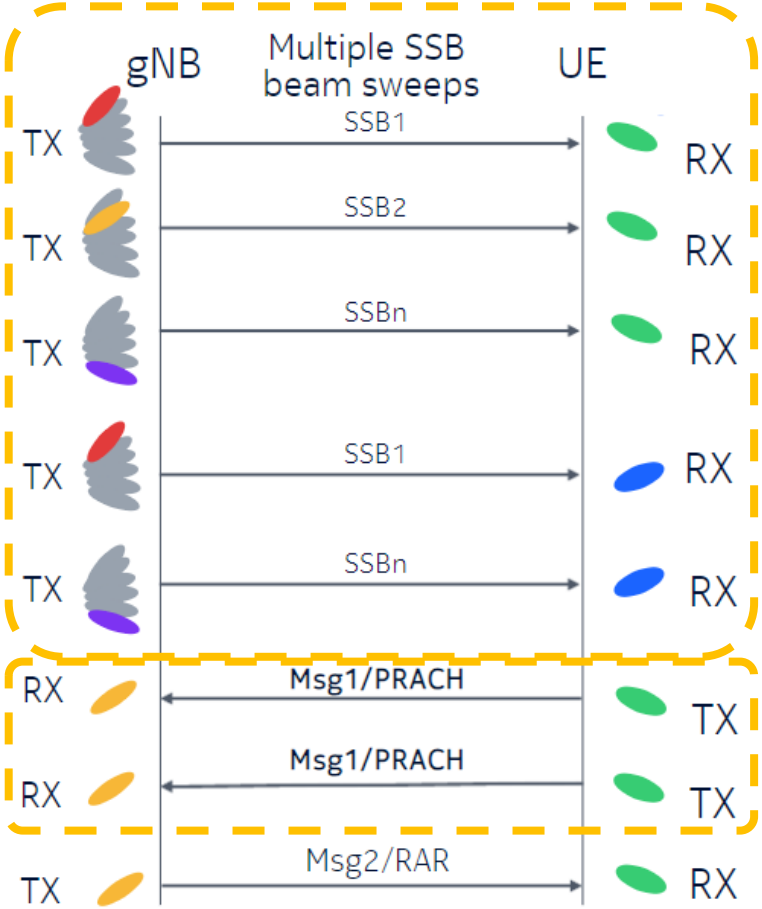
UE is allowed to use transparent techniques to increase transmit power without causing signal degradation, yielding coverage increase.

NR Coverage Enhancements - PRACH Repetitions



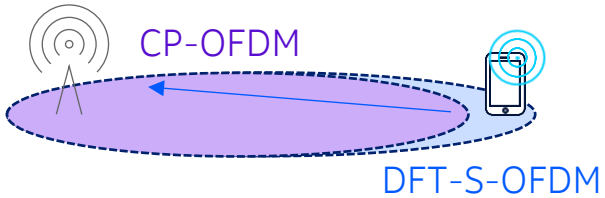
Expected Performance

- 2 to 2.5 dB gain for every repetition number doubling of msg1
- Up to ~5 dB Link budget increase when 4 repetitions are configured [TR 38.830 - Rural deployments in FR1]

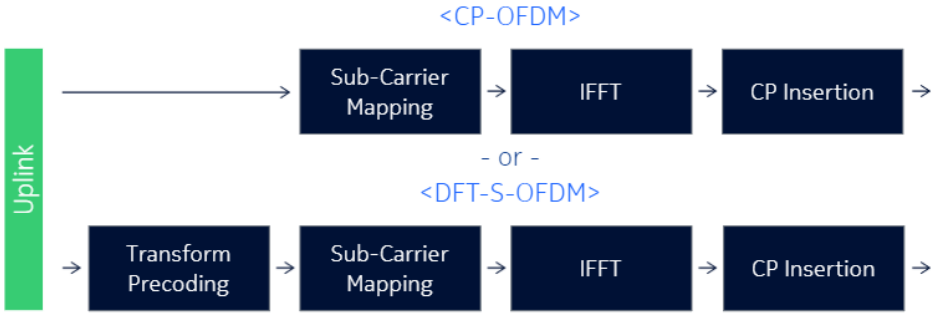


NR Coverage Enhancements - Dynamic waveform switching

Cyclic Prefix Orthogonal Frequency Division Multiplexing



Discrete Fourier Transform Spread Orthogonal Frequency Division Multiplexing



DFT-S-OFDM spreads the data symbol across multiple, contiguous sub-carriers, or tones, like a single carrier transmission, but using OFDM. This results in a lower Peak-to-Average Power Ratio (PAPR) which allows for higher achievable overall signal power and improves the signal to noise ratio.

Benefit of DFT-S-OFDM

- Coverage-enhancing

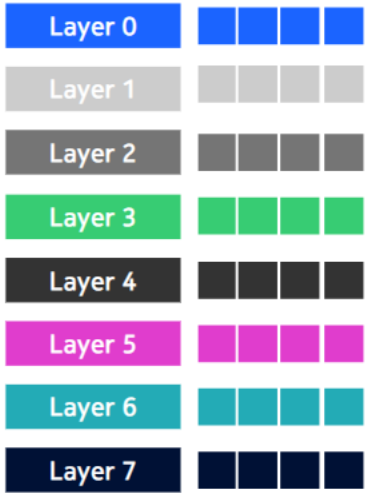
Expected Performance

Transmit power can be increased up to 1 dB thanks to the switch from CP-OFDM to DFT-s-OFDM

Advanced mMIMO

Rel-18 UL-MIMO 8Tx Antenna Configurations targeting CPE / FWA / vehicle / Industrial devices

(M, N, P) per group	Antenna Layout	Antenna Patterns/Antenna Element Gain
(2, 2, 2), (1, 4, 2)		Isotropic (Indoor/Outdoor FWA & Industrial) 8 dBi, 65° HPBW (Outdoor FWA)

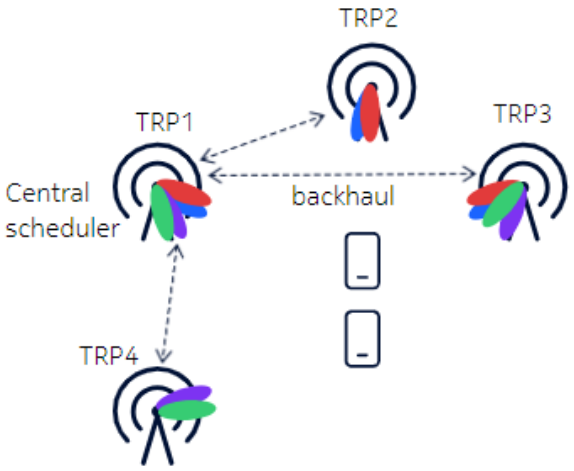


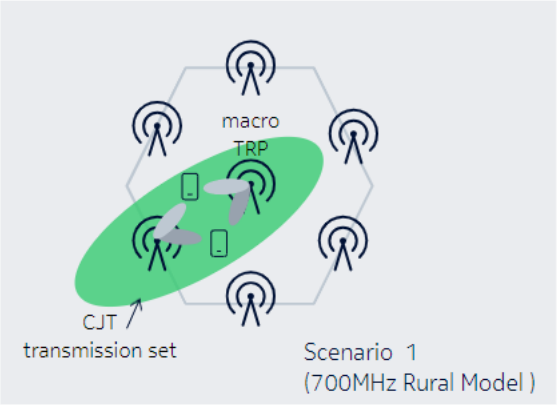
+60%

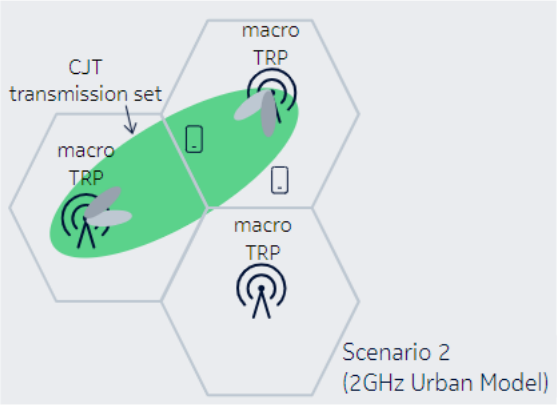
Advanced mMIMO

Coherent Joint Transmission

- **Principle:** each MIMO layer is transmitted by multiple **Transmission Reception Points** (TRPs).
- **Purpose**: coherently align the signals transmitted from multiple TRPs for improved reception at the UE.
- **Mean**: Rel-18 extended the Type II CSI (channel state information) acquisition to support CJT with **up to four TRPs.**







Performances

Significant gains over single TRP transmission at 700MHz Rural Model, lower gains at 2GHz Urban Model

Source: Nokia simulations submitted to 3GPP: 3GPP R1-2205289, R1-2207546, R1-2204540

Mobility Performance In 3GPP Release 18

- Improved mobility robustness
- Improving FR1-FR2 mobility
- Reduced signaling
- Reduced service interruption time

CHO: Conditional HandOver
DC: Dual Connectivity

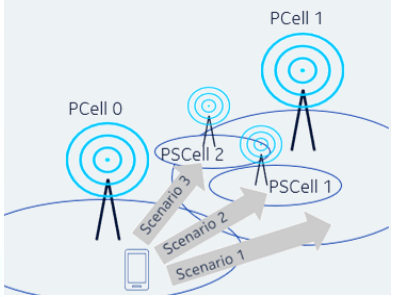
PCell: Primary Cell
PSCell: Primary Secondary Cell

5G Advanced mobility innovations

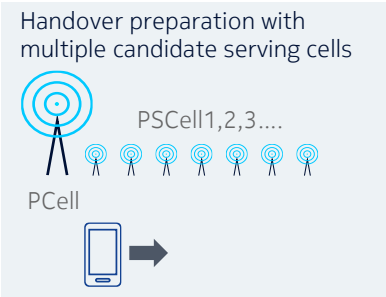
- Throughput enhancements
  - CHO in NR-DC scenario.
  - Subsequent switching of secondary cell group
- Interruption time

  L1/L2 (or Lower layer) Triggered Mobility (LTM)
  - Interruption time significantly reduced
  - Signaling reduced
  - Intra-gNB only in Rel. 18

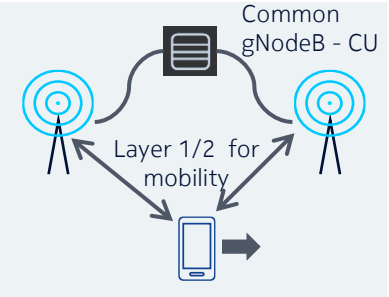
CHO with PSCell quality info



Subsequent switching of secondary cell groups (FR2)



L1/L2 based mobility - inter-cell mobility



Scenario 1 is for single connectivity
Scenario 2 and 3 are for seperate target PSCells

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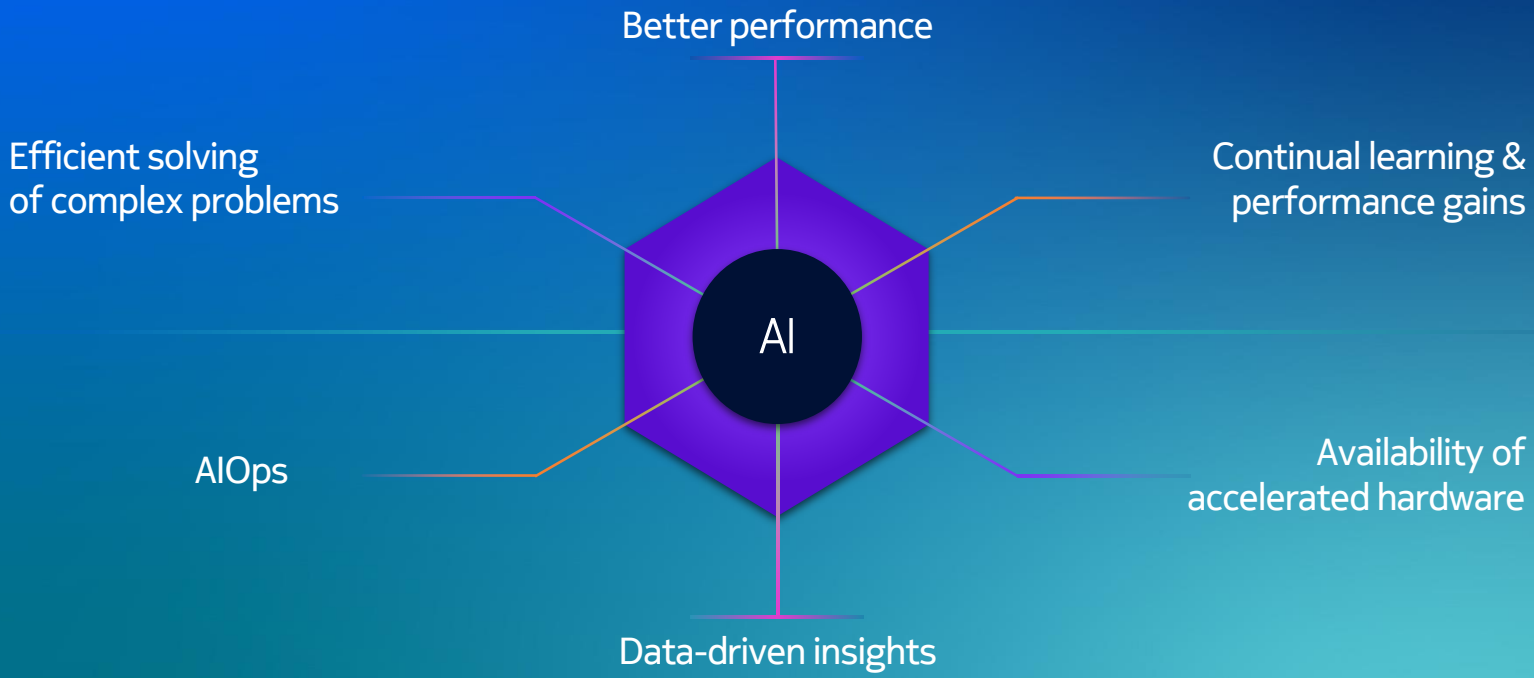
Why AI/ML? What is the AI advantage?

Data accessibility

Cost-efficient accelerators

Break-through of Generative AI.

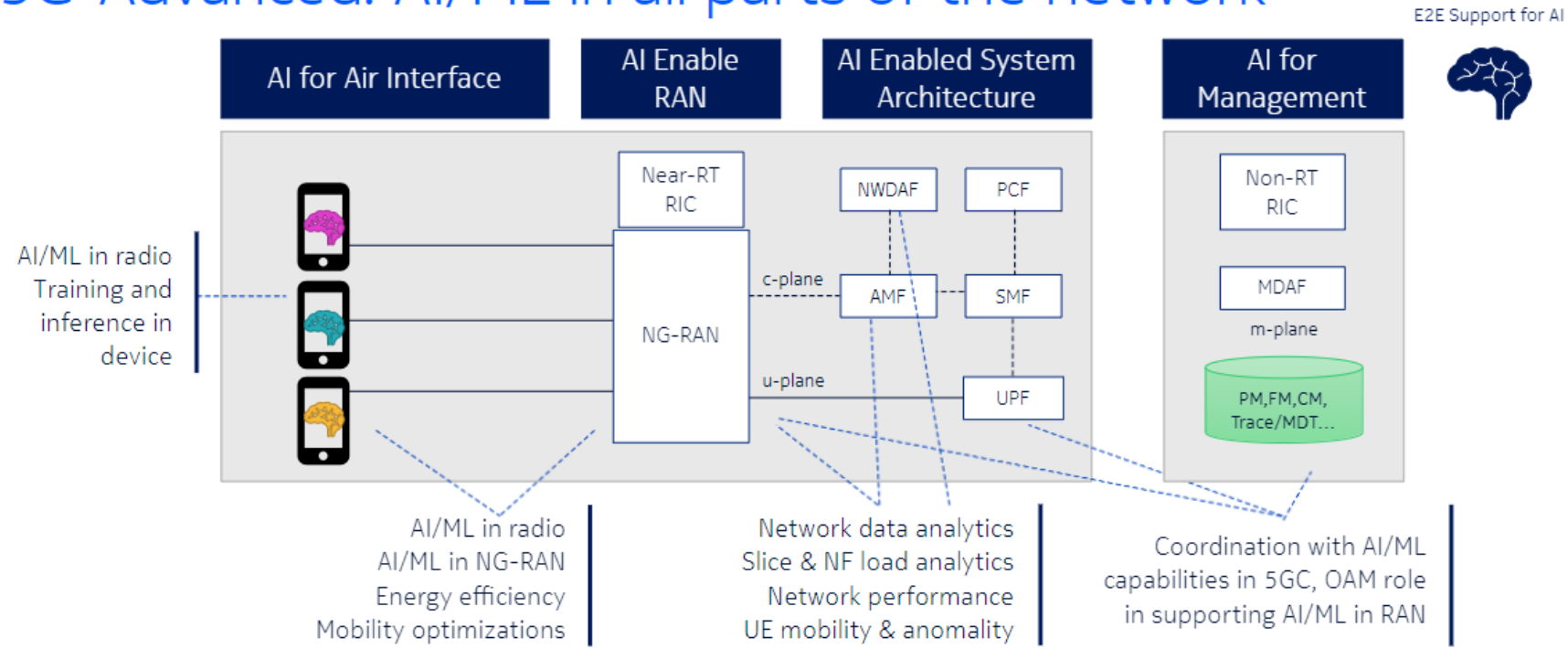


AI/ML timeline in Standardization

5G-Advanced is the foundation of AI/ML in 6G

AIMLsys: system support for AI/ML-based services

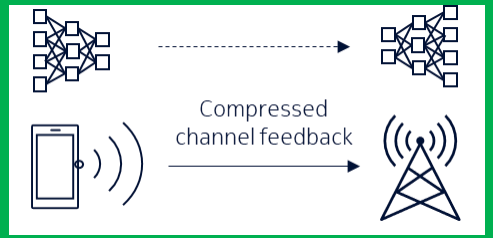
5G-Advanced: AI/ML in all parts of the network



CM: Configuration Management
MDT: Minimizing Drive Tests
RT: Real Time
RIC: RAN Intelligent Controller
MDAF: Management Data Analytics Function
PM: Performance Management
NWDAF: Network Data Analytics Function
FM: Fault Management

AI/ML in air interface: Rel-18 use case families

CSI feedback enhancements

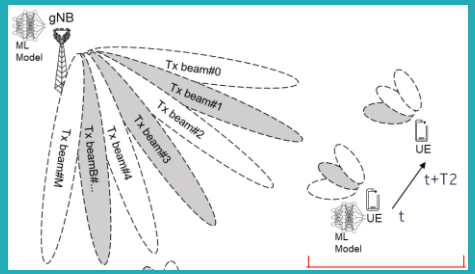


ML-based CSI feedback compression

UE-sided channel prediction

Throughput improvements and overhead reduction

Beam management

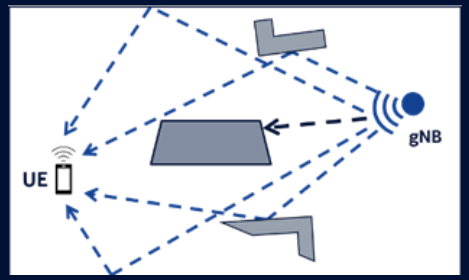


Spatial domain beam prediction

Time domain beam prediction

Overhead, ping-pong and latency reduction

Positioning



NLOS detection

Measurement correction

Positioning accuracy enhancements (cm level)

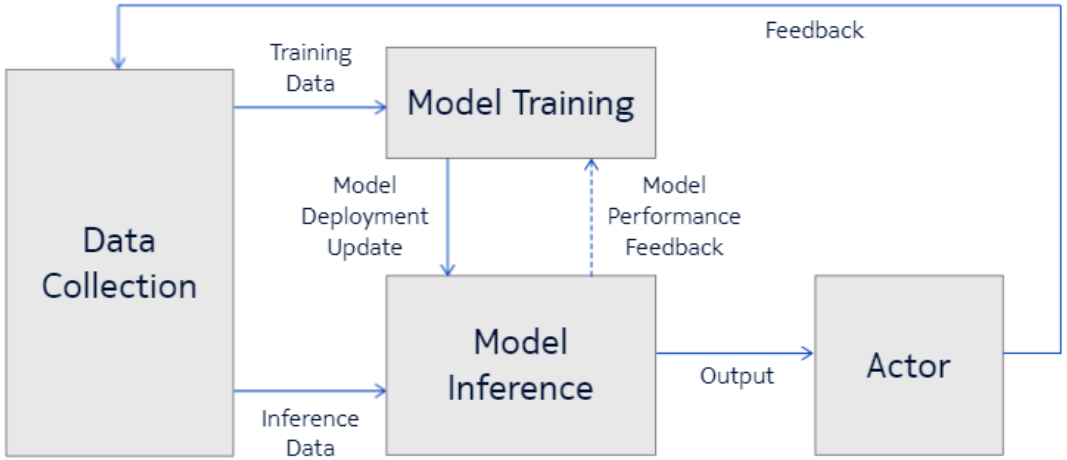
Rel-19: new use cases, e.g. mobility optimization, reinforcement learning…

CSI: Channel State Information

NLOS: Non-Line-Of-Sight

Source: TR 38.843 V1.1.0

AI/ML in NG-RAN: Overview



Specification impact:

- TS 38.300
- TS 38.401
- TS 38.420
- TS 38.423

Data collection procedures:

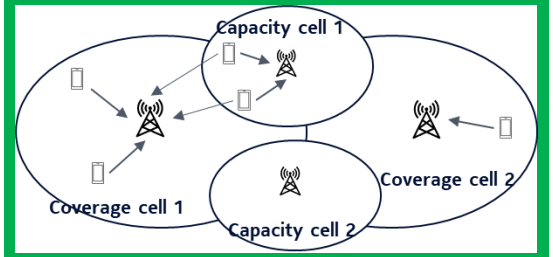
- Predicted resource status information
- UE performance feedback
- Measured UE trajectory
- Energy Cost

- **Model training** located in OAM or in NG-RAN node
- **Model inference** with use-case specific output in NG-RAN Node
- **Actor** triggers or performs corresponding actions
- **Feedback** for ML model performance monitoring

OAM: Operation Administration and Maintenance

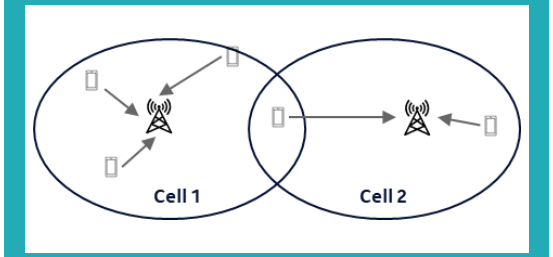
AI/ML for NG-RAN use cases

Energy savings



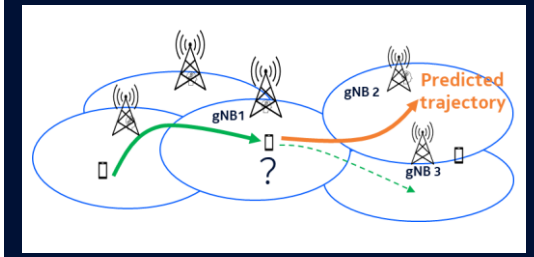
- Switch-off/on of cells based on load and UE trajectory predictions
- Exchange of energy efficiency metrics over Xn interface (Energy Cost IE)

Load Balancing



- Hand-off UEs from highly loaded to lightly loaded cells
- Prediction of:
  - own, neighbor cell load
  - UEs to be handed over to target node

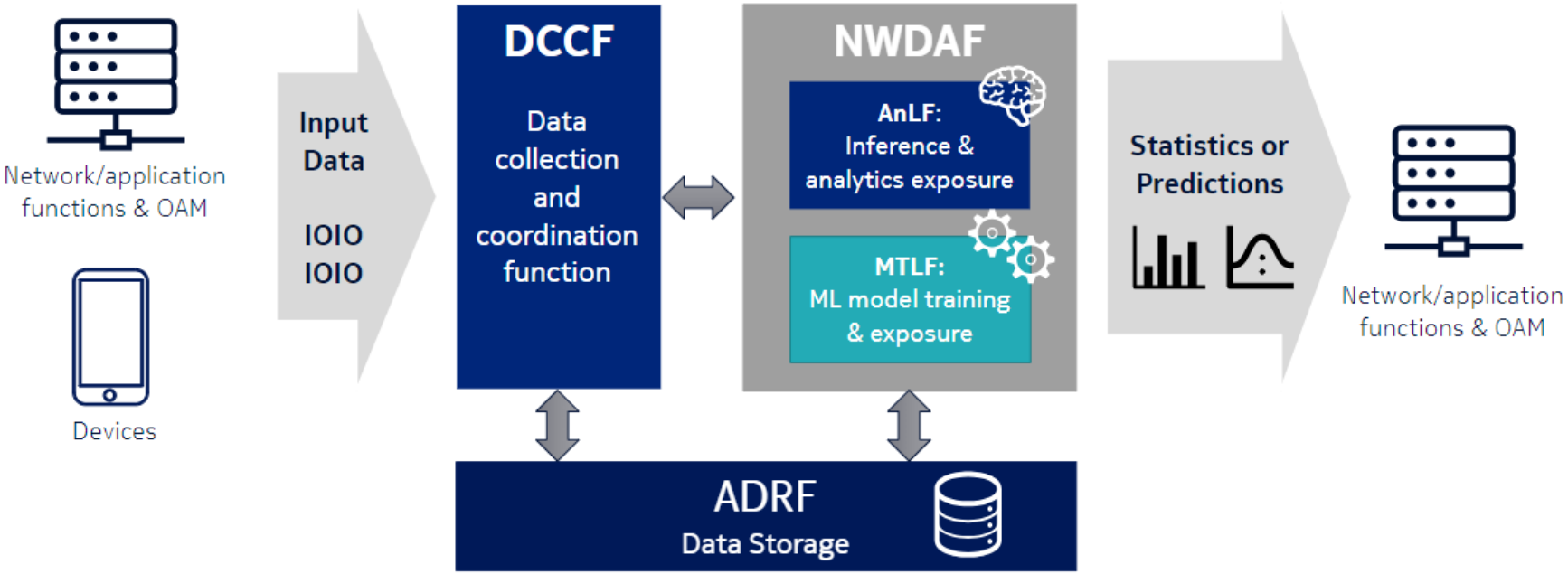
Mobility



- Reduce number of unintended events during mobility (too late, too early, wrong cell preparation)
- UE trajectory prediction on cell-level granularity
- UE performance feedback for throughput, packet error rate, packet delay

Rel-19: Further enhancements for energy savings, mobility, data collection possible

AI/ML for 5GC - NWDAF



DCCF: Data Collection Coordination Function
ADRF: Analytics Data Repository Function

AnLF: Analytics Logical Function
MTLF: Model Training Logical Function

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Extended Reality and the Metaverse

Virtual Reality (VR)
A digital environment that can be either simulated or a replica environment

Mixed Reality (MR) – video see through
A mixture of real and virtual environments - evolution from VR to AR

Augmented Reality (AR) – optical see through
Digital (3D) information overlayed on real images viewed through the device

A true **Metaverse** includes

- **Fully immersive space**
- **Multi-user environment**
- **professional and user generated content**

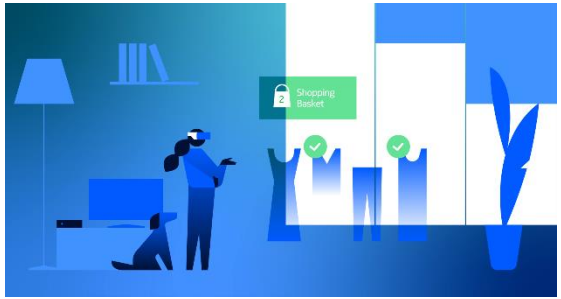
> **“The metaverse is an immersive digital experience that connects people, places (including 3D virtual spaces), things, and/or information in real time in a way that enhances the physical world.”**

Addressing all aspects of life **(gaming, e-commerce, education, etc.)**

XR Use Cases

Social VR

VR spatial shopping



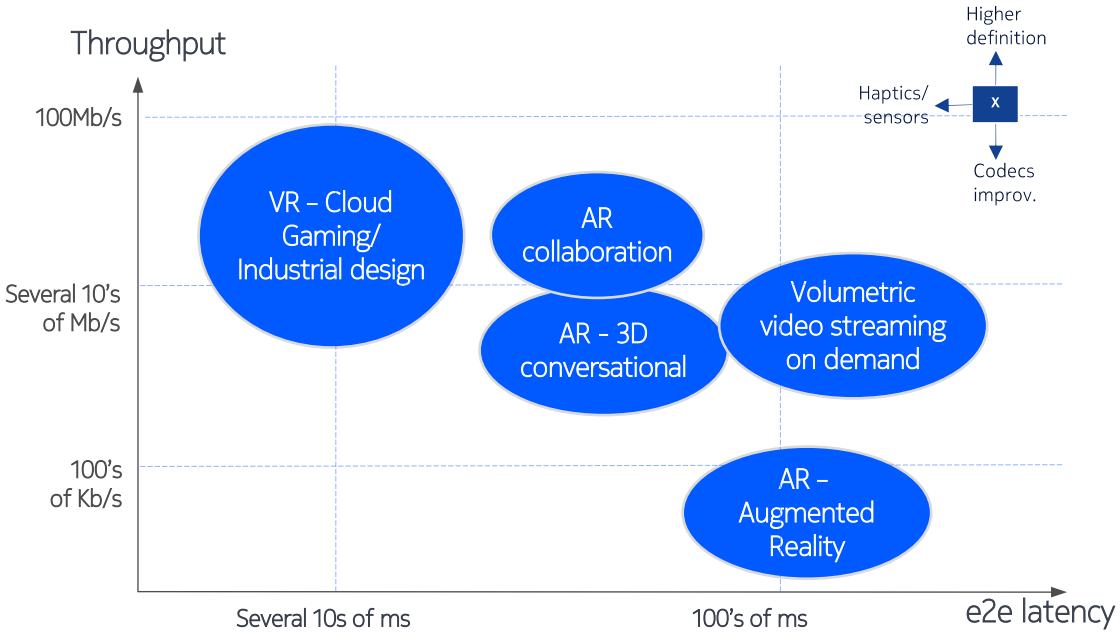
Enterprise VR collaboration

Augmented sports

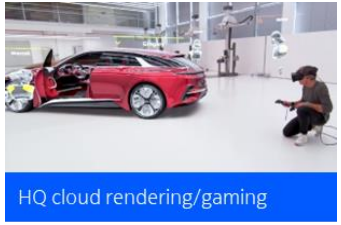
Holo TV (Volumetric video)

Industrial worker assistance

Use Cases and Network Requirements



Enterprise and Consumer XR:
VR Cloud Rendering (photorealistic industrial design), VR Cloud Gaming


HQ cloud rendering/gaming

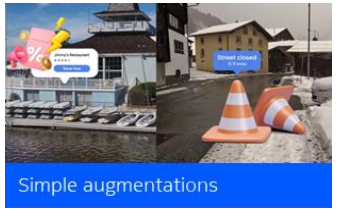
Enterprise and Consumer XR:
AR 3D Holograms, Avatars , Collaboration, Conversation

Multi-user collaboration

VR / AR: 3D video streaming on demand,
e.g., News, Music

Volumetric video

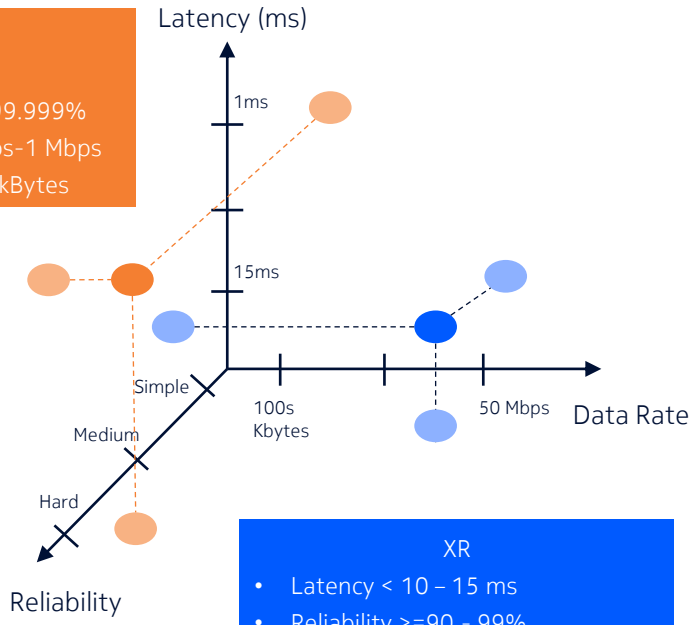
AR: Navigation, Notification


Simple augmentations

Required **throughput** depends on video resolution
Low latency due to multi-user aspects and/or Motion-to-Photon (MTP) latency

XR Traffic Characteristics

IIoT
- Latency < 1ms
- Reliability up to 99.999%
- Data Rate ~ 1kbps-1 Mbps
- Burst size ~ 100 kBytes



XR
- Latency < 10 – 15 ms
- Reliability >=90 - 99%
- Data Rate > = 30-45 Mbps
- Burst size ~ 50 - 100 kBytes
- Period(1/30, 1/60, 1/90, 1/120)s

- Quasi-periodic traffic in large chunks
- Irregular intervals and variable size (in the order of 3x)
- High data rate, including UL for AR services
- Simultaneous transmission of 3D video stream and control data over the same e2e connection
- Traffic is carried over SRTP/RTP or other application protocols such as QUIC.
- Video frame rates:
  - For XR applications, minimum 60fps is the accepted frame rate but 90fps offers better user experience. (For 2D applications, minimum 30fps is the widely accepted)

RTP: Real-time Transport Protocol

SRTP: Secured RTP

QUIC: Quick UDP Internet Connections

XR devices, today and expected

Virtual Reality

VR Headsets

Meta Quest 2

Pico Neo 3

- Wifi standalone
- WiFi or cable PC tethered

VR Metaverse

Now

Augmented Reality (Mixed Reality)

Smartphones

- Wifi
- 5G

Glimpse of AR Metaverse

Now

Smart Glasses XR Viewers

Meta Ray-ban Stories

Xreal Air

- Cable, Bluetooth, WiFi tethered to 5G smartphone

Glimpse of form factor, but no AR Metaverse

2022+

Video see-through MR Headsets

Meta Quest Pro

Apple Vision Pro

- WiFi standalone
- WiFi or cable PC tethered

AR Metaverse in impractical form factor

2023+

Optical see-through AR glasses – 1st Gen

Qualcomm reference design

Qualcomm

Goertek

- WiFi tethered to 5G smartphone

AR Metaverse

2025+

Optical see-through AR glasses – 2nd Gen

Expectation

SAMSUNG

- WiFi tethered to 5G smartphone
- 5G standalone with integrated modem

AR Metaverse

2030+

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Why NTN?

LEO satellites disrupt space

- Satellite cost reduced
- Launch cost reduced
- Launch intervals reduced
- #satellites per launch increased
- End-to-end latency reduced
- Throughput increased

NTN applies to many use cases

TN backhaul	Fixed wireless access	Mobile users	Internet of things
High gain antennas, potentially in mmWave spectrum	Large antennas, no user mobility	Handheld devices, cars	Low-data rate services with long battery life target
AT&T - OneWeb KDDI – Starlink Verizon – Amazon	Starlink, Amazon, OneWeb, Eutelsat, Viasat	AST, Lynk, Starlink, Apple, Qualcomm, MediaTek 3GPP rel 17, 18	Eutelsat and many, many others

Plus NTN provides resilience

LEO: Low Earth Orbit

Satellite communication is evolving from proprietary technologies to 3GPP standard based solutions (1/2)

2020- Proprietary

FWA
Mobile Backhaul

STARLINK
OneWeb

IoT

2023/2024- Leveraging 3GPP for Direct-to-Device

Standard 3GPP handset access enabled by custom NTN tech

AST SpaceMobile
LYNK
STARLINK

Proprietary chipset in 3GPP handset to support satellite spectrum

Globalstar

2024/2025- 3GPP Transparent

Direct-to-Device, VSAT

NR

omnispace

IoT

SATELIOT
skylo
ligado NETWORKS
inmarsat

Chipsets/devices

MEDIATEK
SAMSUNG
Qualcomm

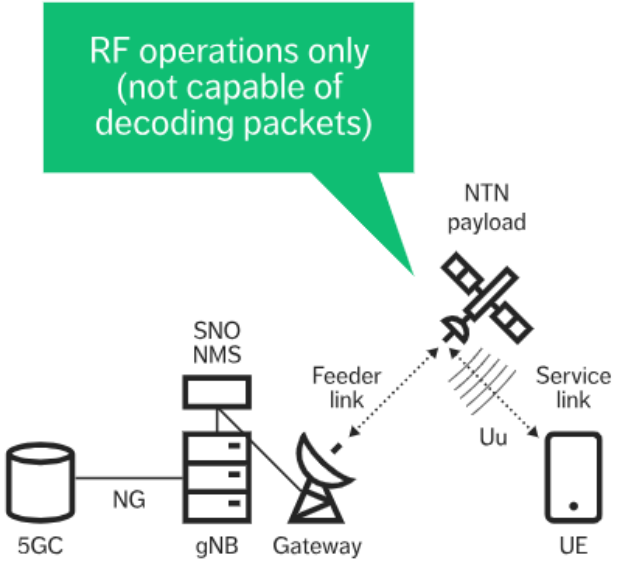
2026+ 3GPP Regenerative

Direct-to-Device, VSAT

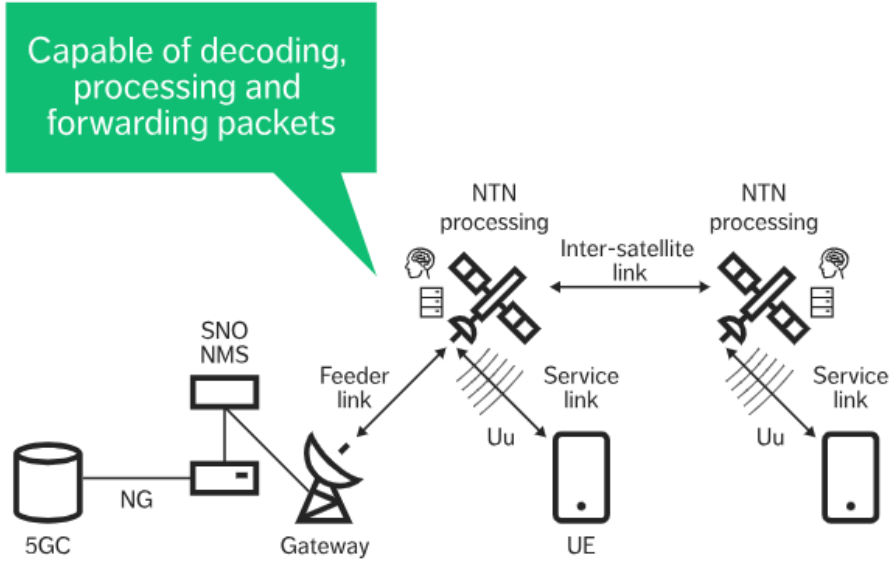
OneWeb
Gen 2

VSAT: Very Small Aperture Terminal

Satellite communication is evolving from proprietary technologies to 3GPP standard based solutions (2/2)

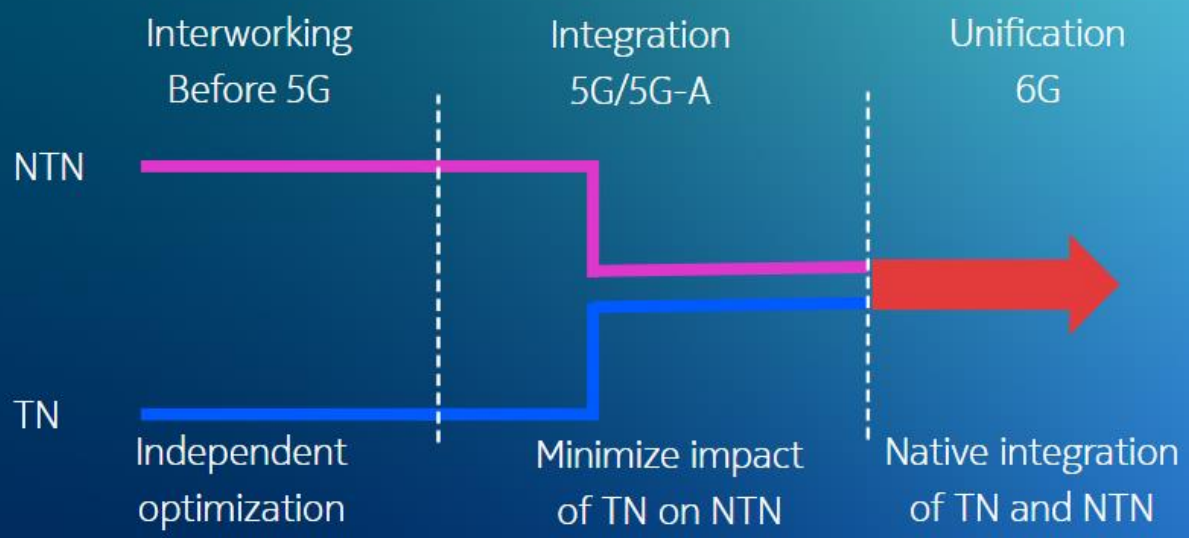


Transparent architecture



Regenerative architecture

Convergence between Terrestrial and Non-Terrestrial Networks



3GPP Standard protocol

Ubiquity, continuity and resiliency

UAS in 5G-Advanced

3GPP specifications for Uncrewed Aerial System (UAS) enable:

- Interoperability and scalability
- Public and non-pubic (private) network solutions
- Spectrum for reliable communications

Unified identification, authentication and authorization

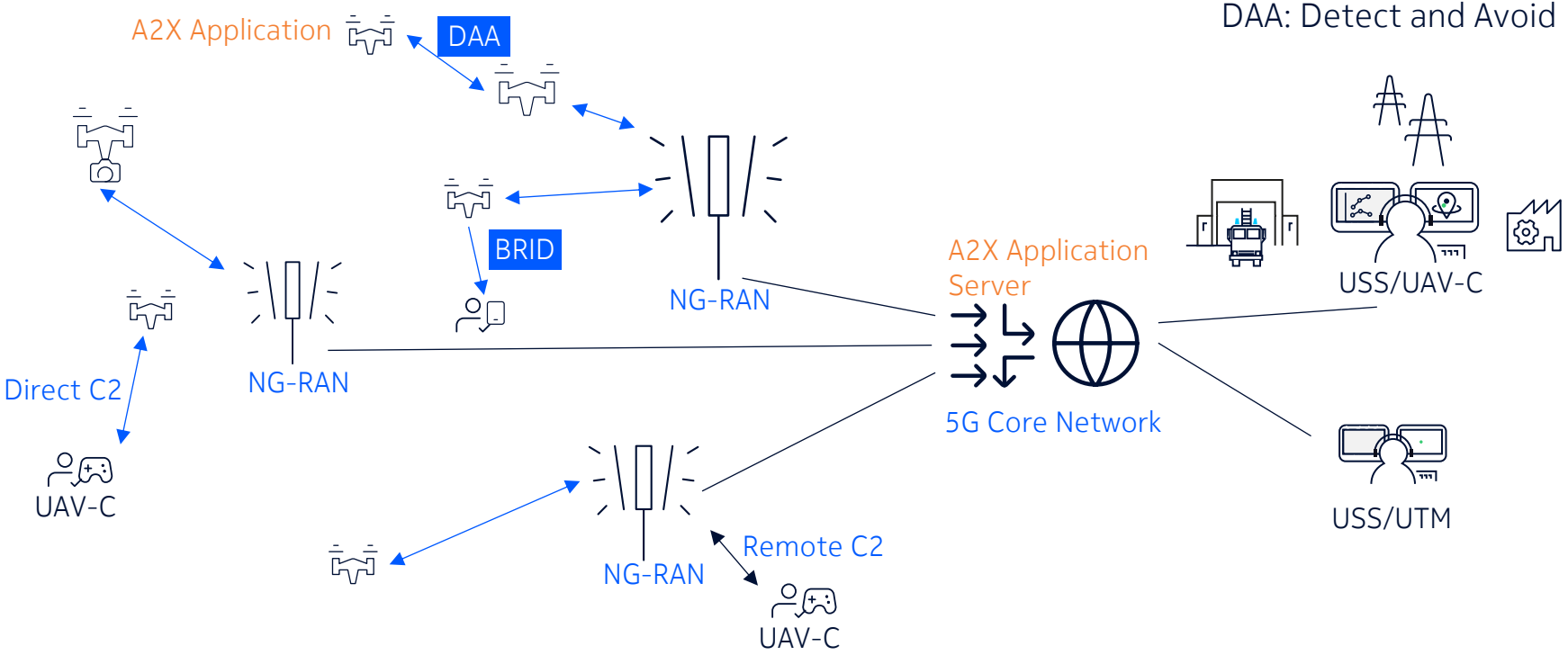
Regional, geographical and commercial conditions

Efficiency and co-existence

Aerial Vehicle capabilities in 3GPP

1. A2X Service

A2X: Aircraft-to-Everything
USS: UAS Service Supplier
BRID: Broadcast Remote ID
DAA: Detect and Avoid

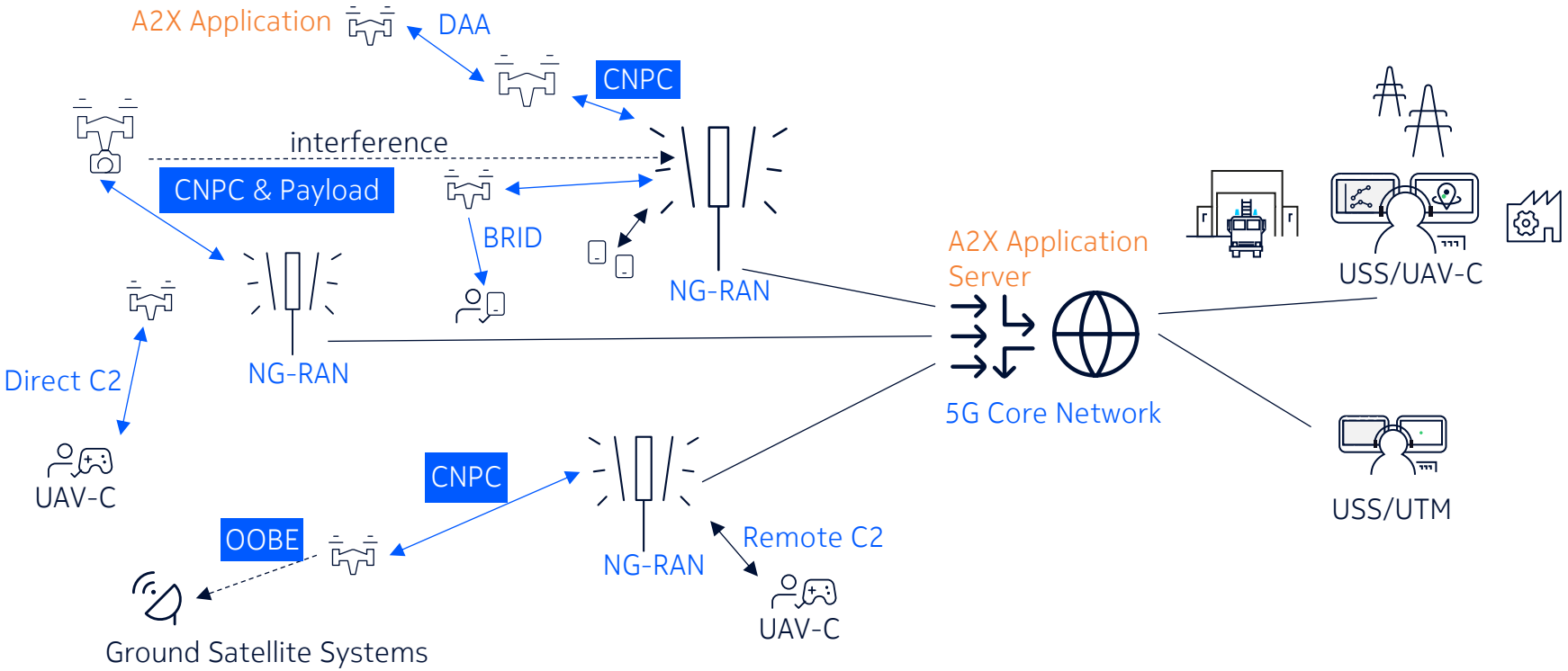


Aerial Vehicle capabilities in 3GPP

2. Reliable, secure and long-distance flight control

CNPC: Control and Non-Payload Communications

OOBE: Out-of-Band Emissions

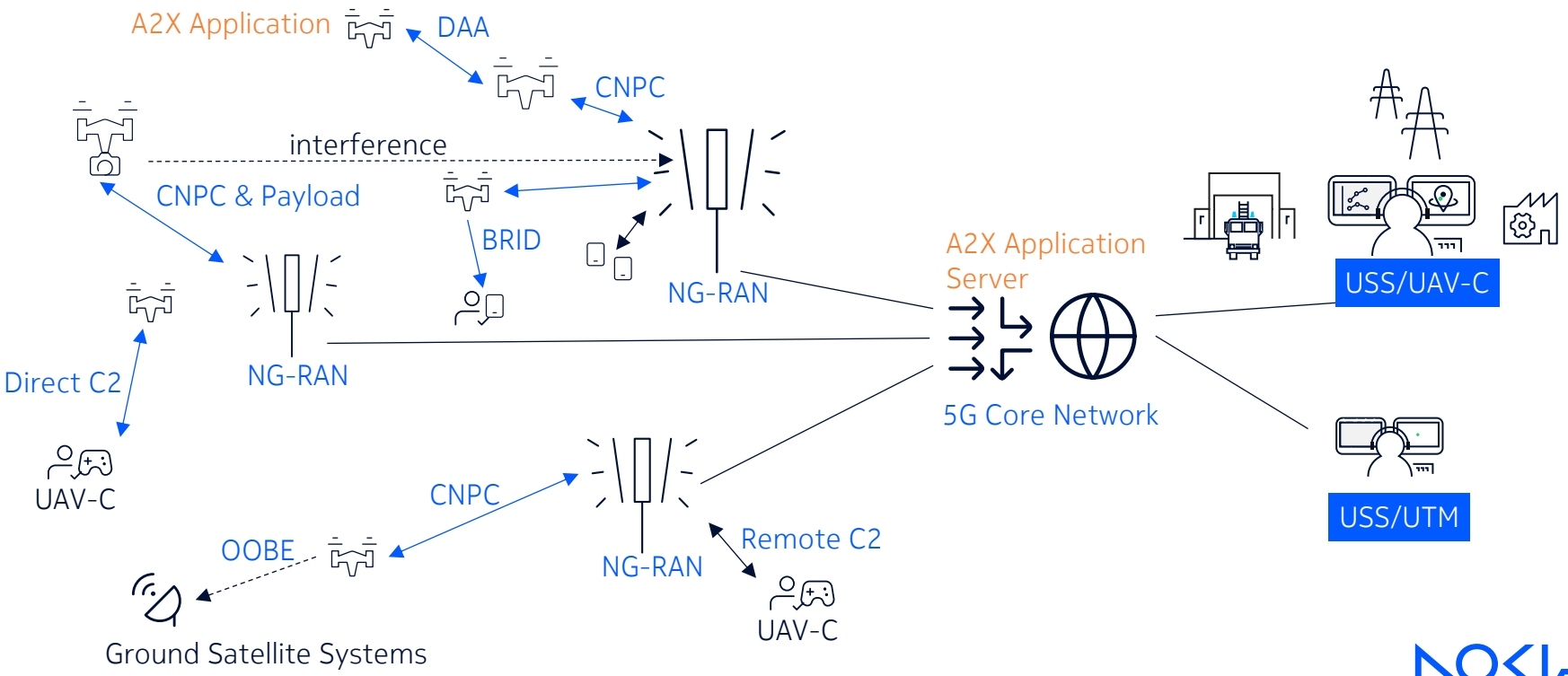


Aerial Vehicle capabilities in 3GPP

3. Multi-UTM deployments support

UTM: UAV Traffic Management

USS: UAS Service Supplier



Evolution to 5G-Advanced

Module Structure

1. 5G-Advanced Key Areas
2. Radio Performance Boosters
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6. Superaccurate Positioning
7. RedCap for IoT
8. Energy Efficiency

Positioning evolution
Towards less human intervention

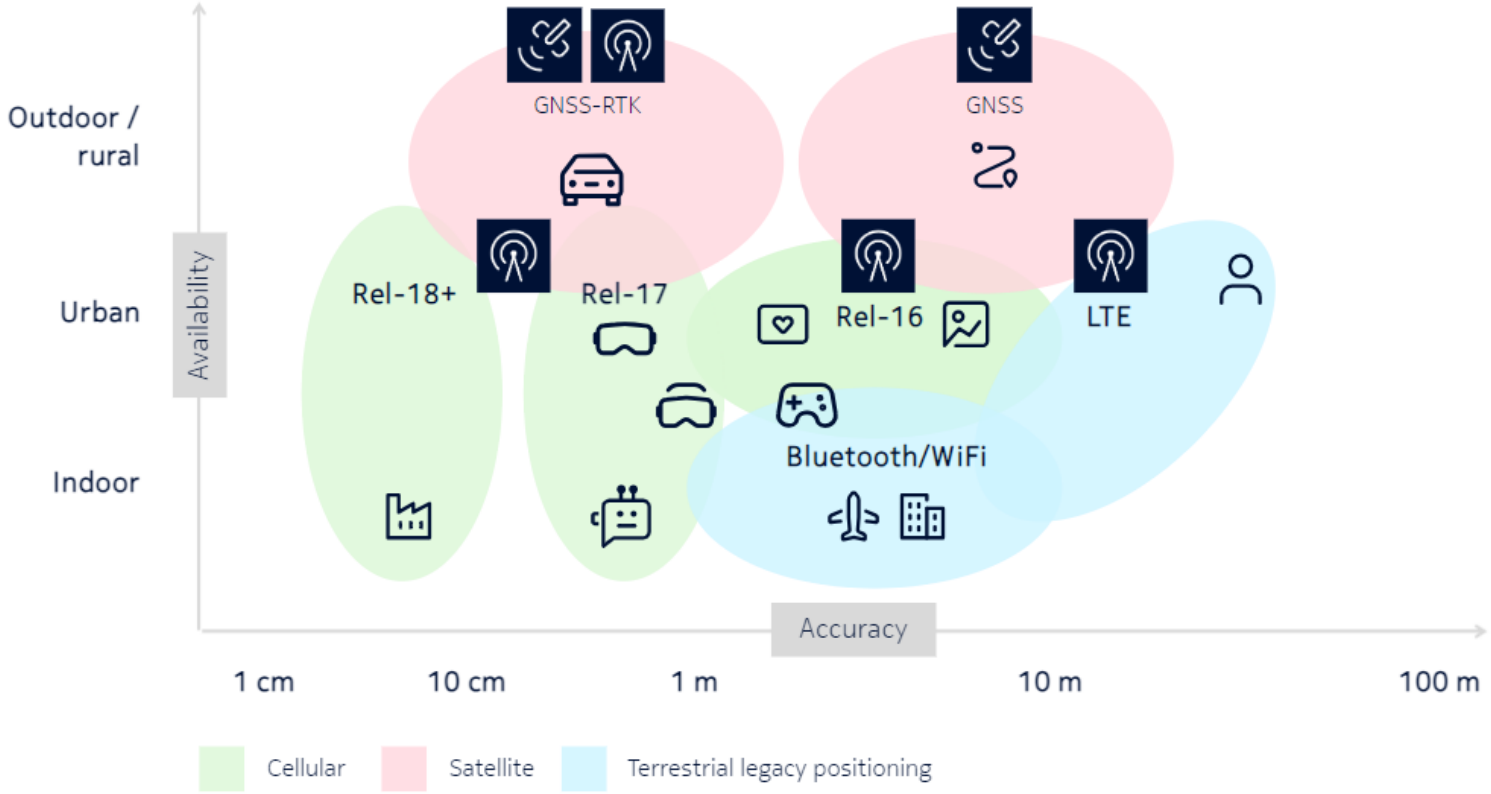
5G Advanced builds in additional techniques that bring accuracy down to cm level.

4G positioning was designed for mandatory emergency services (911), where human intervention is needed. Accuracy requirements of ~tens of meters are sufficient.

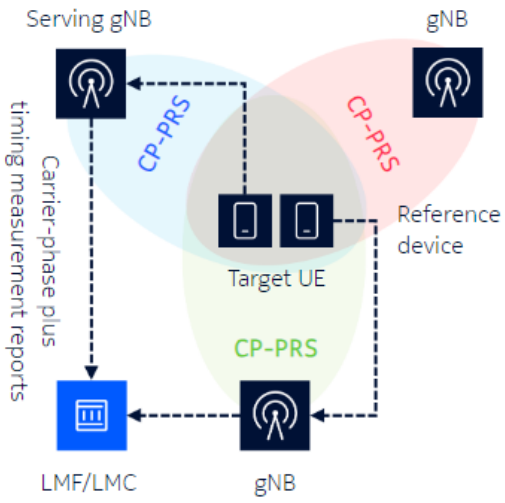
Driven by commercial and industrial use cases, 5G positioning enables high accuracy without human intervention.

Positioning Solutions and Use Cases

GNSS: Global Navigation Satellite System
RTK: Real Time Kinematic



Positioning in 3GPP Release 18 and beyond



Carrier phase positioning, bandwidth aggregationa

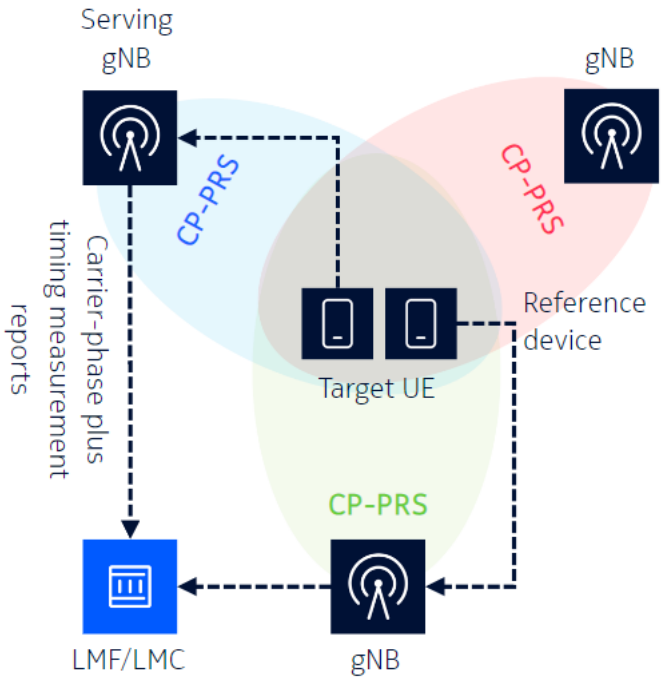
Artificial Intelligence based positioning

Sidelink Positioning

Carrier Phase-based Positioning

Motivation and overview

- Increase accuracy by a factor of ~10 compared to purely time-based positioning methods
- 5G NR based solution can work indoors and outdoors (Provide an indoor complement to outdoor GNSS-RTK)
- Enable new use cases such as high precision tracking of objects and XR in industrial environments



CP-PRS: Carrier Phase-Positioning Reference Signal

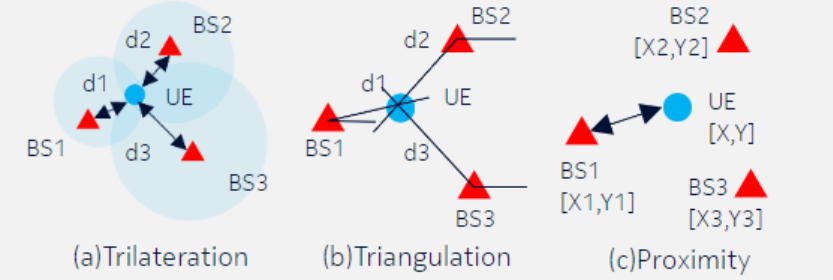
LMF: Location Management Function

LMC: Location Management Component

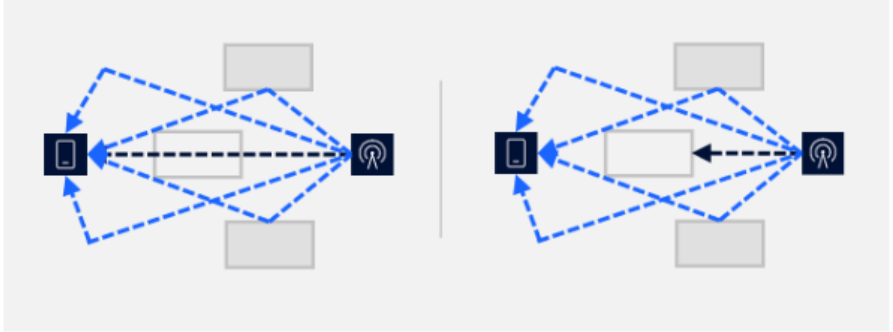
Why AI/ML Positioning?

AI/ML Positioning	• Leveraging AI/ML models to estimate the UE position • Purpose: Improve accuracy

Positioning: Legacy (Rel16->Rel-18)



Positioning: AI/ML Positioning (Rel18 Study Item)



Non-ML approaches:
Positioning methods: Time, angle, phase-based measurements such as CPP (Carrier phase positioning), A-GNSS, OTDOA (Observed Time Difference of Arrival), E-CID (Enhanced Cell-ID), Barometric, WLAN/Bluetooth based

NLOS multipath channel (a) Direct path is attenuated but detectable (b) Direct path is completely blocked or non-detectable

Positioning enhancement in heavy NLOS conditions

CPU | GPU | NPU | RAM | Parallelism | ……

CPU: Central Processing Unit
GPU: Graphic Processing Unit
NPU: Neural Processing Unit
RAM: Random Access Memory

Implementation of AI/ML Positioning

AI/ML Positioning Approaches	
Direct AI/ML	The output of AI/ML model inference is UE location.
AI/ML assisted	The output of AI/ML model inference is new measurement and/or enhancement of existing measurement.

AI/ML model deployment option			
One-sided	UE-side model	gNB-side model	LMF-side model

AI/ML model - Supervised Learning

Data collection

Model Training
Model Inference
Model Monitoring

Performance targets and KPIs

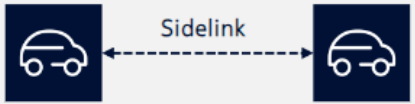
Accuracy
Complexity

Sidelink (SL) Positioning

What is sidelink

Direct link between UEs (as opposed to UL/DL between UE and gNB).

Why sidelink

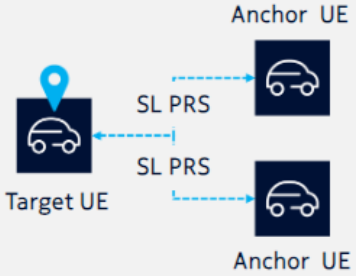


To determine:

- **Absolute position or velocity** (not necessarily)
- **Relative position** (to other network elements or UEs)
- **Relative velocity** (to another UE)
- **Range** (distance and/or angle between UEs), in addition to

To address **recent requirements** of **IIoT, V2X, public safety, commercial** use cases

What's new on sidelink



New reference signals **SL PRS** transmitted between target UE and anchor UEs for **Time, angle, and power** measurements

New protocol **SL Positioning Protocol (SLPP)** for coordination between UEs

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IoT Technology

Critical IoT

Time-critical data delivery with strict delay constraint and high reliability

Broadband IoT

High data rates and lower delay typical of mobile broadband services

Massive IoT

Delay-tolerant, low data rates low-cost devices, low energy consumption

Ambient IoT

Ultra-low complexity devices, ultra-low power consumption, battery-less or limited energy storage (no manual recharge)

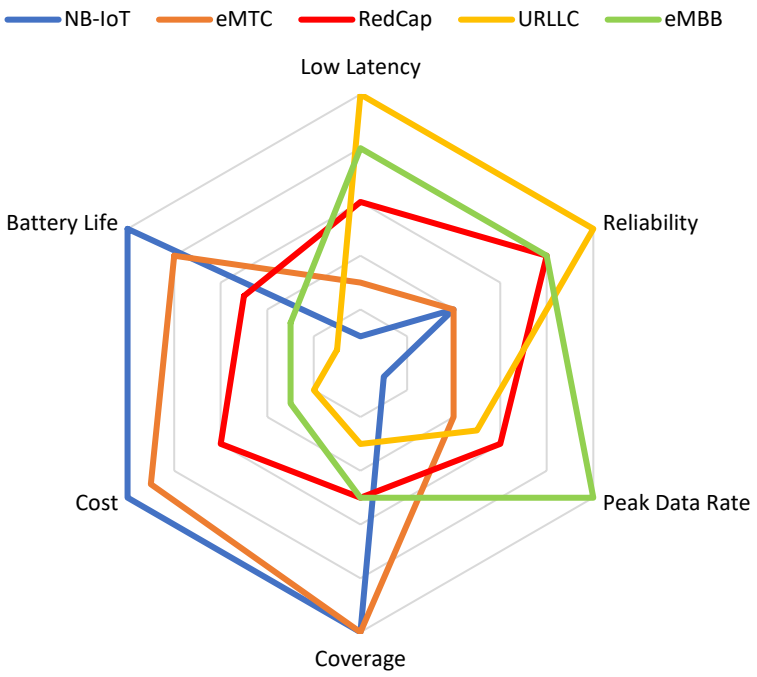
Release 17 RedCap Use Cases

Mid-range IoT

Video
Latency < 500ms, data rates of 2-4 Mbps or 7.5-25 Mbps (HD)

Industrial wireless sensors
End-to-end latency < 100 ms for normal data, 5-10 ms for safety sensors, data rate < 2 Mbps

Wearables
Data rates of 5-50 Mbps in DL, 2-5 Mbps in UL, peak rates up to 150 Mbps DL, 50 Mbps UL



R18 RedCap evolution requirements and work item summary

To further expand the market for RedCap use cases, further complexity reduction and energy efficiency enhancements should be specified.

- Rel-18 RedCap should provide NR support for low-tier devices between existing LPWA UEs and the capabilities of Rel-17 RedCap UEs.

Complexity/cost reduction

- UE BB bandwidth reduction
  - 5 MHz BB bandwidth only for PDSCH (for unicast) and PUSCH, with 20 MHz RF bandwidth for UL and DL
- UE peak data rate reduction
  - The peak rate target is 10 Mbps regardless of what optional features the UE may support.
- Support additional separate early indication(s)

Power saving/energy efficiency enhancements

- Enhanced eDRX in RRC_INACTIVE (>10.24 s)

IoT Technology Comparison

	Massive IoT				Mid-range IoT			Critical IoT
	NB-IoT		LTE-M		eRedCap	RedCap		URLLC
	Cat-NB1	Cat-NB2	Cat-M1	Cat-M2	FR1	FR1	FR2	
Downlink peak rate	27 kbps	127 kbps	300 kbps	2.4 Mbps	10 Mbps	89 Mbps	213 Mbps	
Uplink peak rate	62 kbps	159 kbps	475 kbps	2.6 Mbps	10 Mbps	91 Mbps	228 Mbps	
UE bandwidth	200 kHz	200 kHz	1.4 MHz	5 MHz	20 MHz	20 MHz	100 MHz	
Reliability	99% - 99.9%				99.99%			99.999%
Latency	Up to 10 s for successful delivery of an application layer packet from the device to RAN				5-10 ms safety sensor, 100 ms others User plane latency			1 ms User plane latency
Coverage (maximum coupling loss)	164 dB				144 dB			
Maximum cell size	120 km		100 km		100 km			
Battery life	10 years							
UE power class	14/20/23 dBm				23 dBm			
Frequency range	FR1				FR1	FR1, FR2		

Introduction

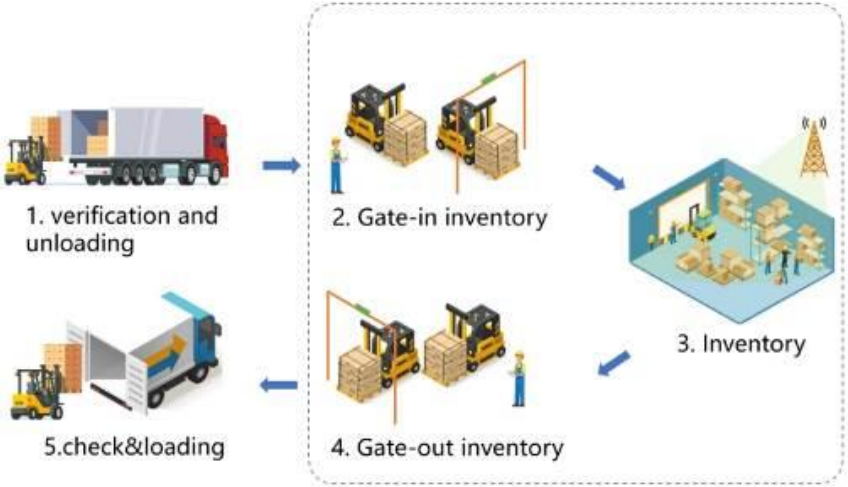
Ambient IoT use cases

SA1 has defined numerous use cases for Ambient IoT

These use cases includes:

- Automated warehouse inventory
- Medical instruments inventory management and positioning

  Intralogistics in automobile manufacturing
- Ambient IoT sensors in smart home
- Ambient IoT for airport terminal / shipping port
- Finding Remote Lost Items
- Indoor positioning in shopping center using Ambient IoT
- and many more …

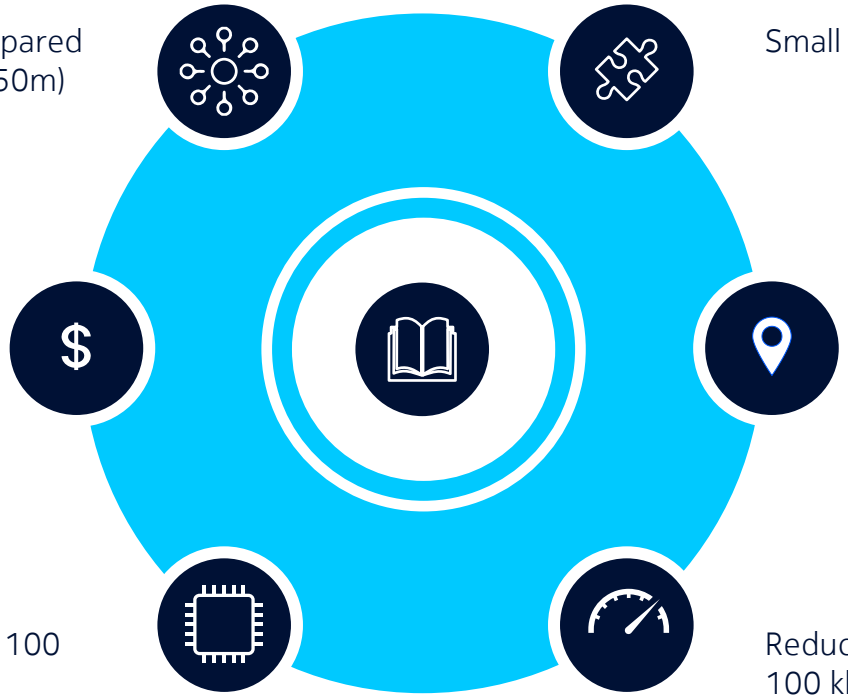


Ambient IoT Technology: Design Targets

Improved range as compared to RFID solutions (e.g., 50m)

Small device size, form-factor

Ultra-low cost (e.g., $0.02 - $0.5)



Relaxed positioning accuracy (e.g., 3 – 5m)

Ultra-low power (e.g., < 100 micro watts)

Reduced data rate (e.g., 10 – 100 kbps)

No need for battery charging or replacement (enabling low maintenance, long life cycle operation)

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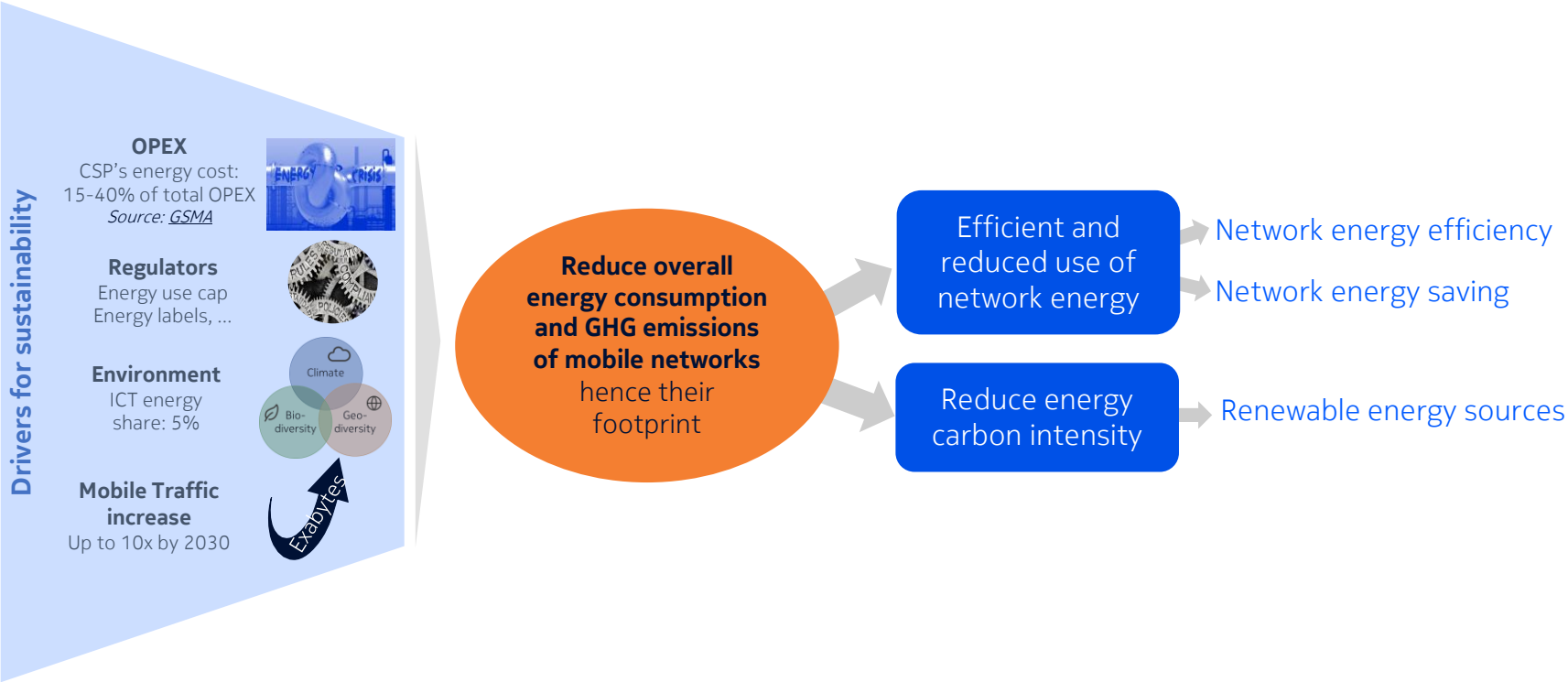
Network energy efficiency and energy saving



CSP: Communication Service Provider
GHG: Green House Gas
ICT: Information and Communication Technology
OPEX: OPerational EXpenditure

Network energy efficiency and energy saving

Network Energy Efficiency

bit/Joule, GB/KWh or Gbps/KW, allows to use less energy per data bit transfer.

It involves measures and technologies to optimize end-user performance while minimizing energy consumption.

Note: Improved energy efficiency does not necessarily result in decreasing the overall energy use.

Network Energy Saving

KWh or saving gain %, allows to save energy also in scenarios with no data bit transfer.

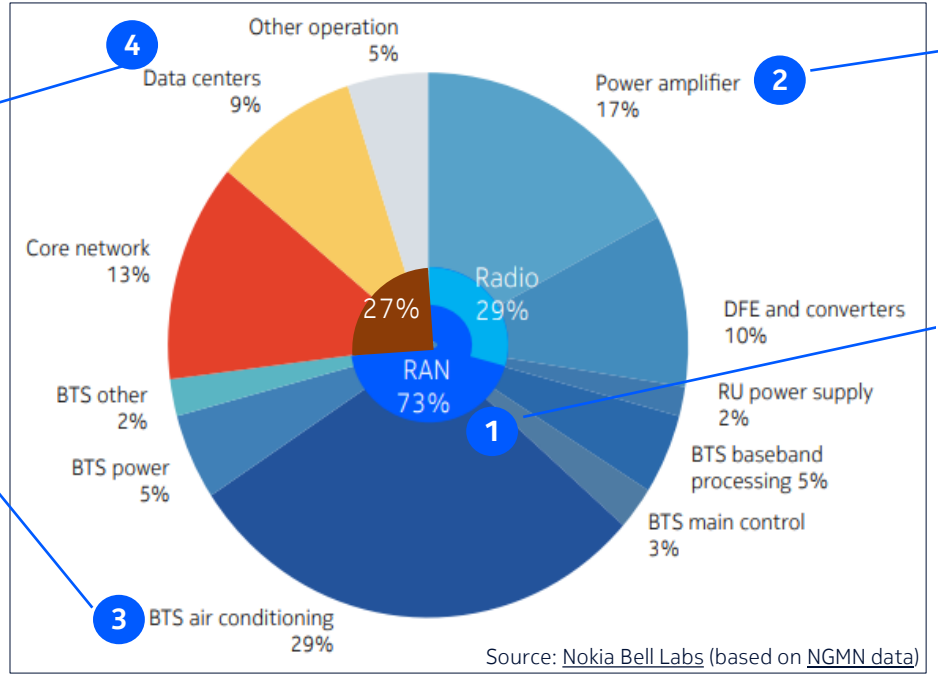
It involves measures and technologies to reduce the overall energy use in the network without compromising QoS.

Mobile network energy consumption breakdown

Data centers energy use largely depends on the operator scenario (excluding the data centers of application providers)

Mobile devices count for ~30% of total network footprint[2], but only 11% of the devices' footprint is due to the operational stage, the rest is due to manufacturing etc. Hence, their contribution to E2E network energy may be omitted

29% share is typical of legacy indoor RAN sites equipped with active air cooling. The share drops to 10% if passive cooling[3] is used and the radio units are placed outdoors (depending on climate conditions for the sites)



Source: Nokia Bell Labs (based on NGMN data)

Within the base station, the power amplifier is the largest contributor to energy consumption and operates with ~40% efficiency.

The RAN has the largest energy contribution in the entire network (73% to 87%[1] based on the source). Thereby, (facilitating) RAN-level approaches targeting to reduce energy use of the base station have the largest energy saving potential

(1) Going green: benchmarking the energy efficiency of mobile networks (second edition)

(2) ABI Research, 'Environmentally sustainable 6G deployments', Sep 2020 Link

(3) Nokia liquid cooling innovations, Link

The 3GPP enablers to optimize mobile network energy consumption are end-to-end

Device	RAN	Transport	Core Network
Device energy saving techniques	Energy-aware radio design, RAN energy saving techniques	Transport energy saving techniques	Energy-aware system architecture functionalities and techniques incl. security & privacy aspects

Energy-aware E2E use cases and service requirements

O&M and automation capabilities incl. data collection/analytics for energy saving

AI/ML capabilities for energy saving

Cloud Infrastructure related energy saving techniques

RAN energy saving techniques enabled in 3GPP Release 18

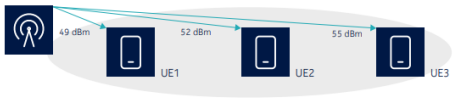
15-30 %

Overall saving in RAN energy consumption*

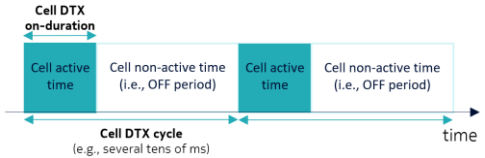
** based on 3GPP BTS power model and evaluation methodology defined in 3GPP TR38.864, in low-to-medium load scenarios with Release 18 UEs*

Antenna adaptation
UE-assisted and dynamic

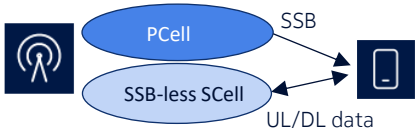
PDSCH power adaptation
UE-assisted and dynamic



Cell DTX/DRX
UE-aware and dynamic



SSB-less SCell
FR1, inter-band carrier aggregation



CSI	Channel Status Information
DTX/DRX	Discontinuous Transmission/Reception
FR1	Frequency Range 1
mMIMO	massive MIMO (Multiple-Input Multiple-Output)
PCell	Primary Cell
PDSCH	Physical Downlink Shared Channel
PRB	Physical Resource Block
SCell	Secondary Cell
SIB	System Information Block
SSB	Synchronization Signal Block
TRS	Tracking Reference Signal
UE	User Equipment

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